Aaradhya Dev Tamrakar

AI Systems & Applied Machine Learning Engineer

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Final-year Electronics & Information Engineering student and Fusemachines AI Fellow. Builds reproducible applied ML, edge AI (ESP32-S3 / TFLite Micro), and formally verified systems. Experienced across PyTorch, C/C++, FastAPI, Docker, and Z3 SMT.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving **100% execution success and answer accuracy** on a 50-question project benchmark.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026 – Present

PyTorch · Python · AIF360 · Fairlearn · FairFace · SHAP · Statistical Testing

- Co-developed an automated fairness-auditing framework with dual **AIF360** and **Fairlearn** backends computing **Disparate Impact Ratio (DIR)**, **Demographic Parity Difference (DPD)**, **Equal Opportunity Difference (EOD)**, and **Equalized Odds**.
- Audited classification performance across **126 granular race × age × gender groups** on the **FairFace benchmark (97,698 images)**, discovering significant disparity drop-offs across intersectional distribution shifts.

SPARK — Safety-Critical Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TFLite Micro · PyTorch · Python · BLE · FastAPI · SHAP

- Technical lead for ML modeling, quantization, and gateway architecture in a four-person team; designed a two-tier fall detection system eliminating cloud latency and privacy hazards.
- Trained 1D-CNN on the **SisFall benchmark (38 subjects, 38,420 windows)**; achieved **0.9185 AUC-ROC (95% CI: [0.9065, 0.9305])** and **87.81% sensitivity** under subject-stratified 5-fold cross-validation with zero data leakage.
- Executed post-training integer quantization (PTQ), compressing model flash memory footprint to **18.5 KB** for deterministic, **sub-100ms** on-device confirmation on the **ESP32-S3** microcontroller.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, PyMC, ArviZ, SHAP, Fairlearn, AIF360, Hugging Face Transformers, OpenCV, MediaPipe

Edge AI & Embedded Systems: ESP32-S3 (ESP-IDF / FreeRTOS), TensorFlow Lite Micro (INT8 PTQ), Arduino, MPU-6050, UART / I2C / SPI, Bluetooth Low Energy (BLE), MATLAB / Simulink

AI Systems, Backend & MLOps: FastAPI, Docker, Model Context Protocol (MCP), asyncio, MLflow, Evidently AI, Airflow, PostgreSQL, SQLite FTS5, REST APIs

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD